

TOULOUSE SCHOOL OF ECONOMICS

Master 1 Thesis

**Understanding 2.5-Degree Price
Discrimination:
Imperfect Signals, Pricing, and Welfare**

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Abstract

This paper examines the economics of 2.5-degree price discrimination in a setting with imperfect signals. We analyze how information design affects pricing strategies and welfare outcomes when firms can commit to nonlinear pricing schemes.

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1 Introduction

Two classical models partition the economics of price discrimination along an information dimension. In second-degree discrimination, the seller knows only the population distribution of consumer types and elicits private information through a nonlinear tariff (Mussa and Rosen, 1978; Maskin and Riley, 1984): the instrument is rich but individual information is absent. In third-degree discrimination, the seller observes a perfect label—a verifiable demographic or geographic tag—and charges group-specific flat rates (Schmalensee, 1981; Varian, 1985): individual information exists but the instrument is blunt. Neither model captures a seller who has *both* an imperfect signal *and* a nonlinear menu—yet this combination increasingly characterizes real markets. Driven by the rapid proliferation of tracking technologies, firms now harvest granular individual-level data from a wide spectrum of sources ranging from web cookies and telematics devices to credit bureaus and respond with differentiated contract schedules rather than posted prices. This paper studies that intermediate case, which we call *2.5-degree price discrimination*, and asks what it does to welfare.

We can see this motivating progression through the evolution of car insurance. Historically, when there was no signal to identify a driver’s underlying risk, the insurer relied entirely on classical screening via a single nonlinear menu. Subsequently, insurers began gathering demographic information and verifiable driving records to better target their pricing strategies. Today, the landscape has shifted further: insurers now receive granular telematics data—acceleration patterns, braking frequency, and travel times—from third-party devices in the form of usage-based insurance scores. These data are informative but imperfect; they sort drivers into rough risk categories without perfectly revealing any individual’s type. Within each category, a high-risk driver still knows more about their own riskiness than the insurer does. To prevent high-risk drivers from claiming the contracts designed for low-risk ones, the insurer must still leave informational rents to low-risk types. This is the standard second-degree distortion, now applied residually after the signal has done its partial sorting. The welfare question is therefore non-obvious: while more information should reduce the screening distortion (potentially re-opening markets for previously excluded consumers), an imperfect signal fundamentally acts as a mean-preserving spread of the virtual type distribution. Crucially, such a spread disrupts the optimal allocation in ways the nonlinear menu cannot fully repair.

This institutional reality naturally raises a critical question regarding information design and data regulation. In practice, the precision of these signals is rarely determined by the principal (the insurer) alone; rather, it is engineered by third-party intermediaries—such as car manufacturers or data brokers—and is increasingly governed by social planners through privacy legislation. To understand how such entities should regulate or design this information structure, we must first unpack its fundamental economic consequences. Specifically, how does the introduction of an imperfect signal impact total social welfare, the firm’s expected profit, and the consumer’s informational surplus? Answering this question, and tracing how the surplus is reallocated among stakeholders, is the central focus of our analysis.

The welfare analysis of price discrimination has a long tradition. Schmalensee (1981) and Varian (1985) establish the foundational result for third-degree discrimination under linear pricing: welfare improves if and only if total output expands. Bergemann et al. (2015) characterize the complete welfare-surplus frontier achievable when a monopolist receives additional consumer information and responds with group-specific linear prices. We study the complementary case: the signal is imperfect and the firm responds with a nonlinear menu rather than a flat rate. This layering of coarse segmentation on top of within-group screening generates a welfare decomposition—expansion dividend versus variance penalty—that has no counterpart in the linear-pricing literature, and our market-shutdown condition is the natural nonlinear analogue of the output-expansion criterion.

Two papers are close in spirit and the contrast with each clarifies our contribution. Bergemann et al. (2022) study an environment in which the *seller* designs the signal received by buyers about their own valuations, asking which information structure maximizes revenue. We study the opposite direction: the signal is generated externally and observed by the *principal* about the agent’s type, so the welfare question shifts from optimal signal design to the welfare consequences of an exogenous mean-preserving spread. Eső and Szentes (2007) show that in an auction the revenue-maximizing seller should disclose all available information to bidders. Our setting differs along two dimensions: the mechanism is nonlinear pricing rather than an auction, and the information flows to the principal rather than from it. These differences overturn the monotonicity result—disclosure can reduce total welfare when no consumer is excluded under the prior, a possibility that cannot arise in their framework.

We study a principal-agent screening model in which a risk-neutral agent has a privately-known type $\theta \sim F$ on a bounded interval. An exogenous information structure generates a signal σ that is publicly observed by the principal, who offers a menu of contracts tailored to the realized posterior. The key analytical object is the *posterior virtual valuation*—the signal-adjusted version of Myerson (1981)’s virtual type—which governs the post-signal screening problem. Two discipline conditions are imposed: the Monotone Likelihood Ratio Property (MLRP), ensuring higher signals predict higher types, and a log-concavity condition on the posterior that rules out bunching.

Our first result is a martingale property: for every type θ , the prior virtual value equals the expectation of the posterior virtual value across signal realizations. Information disclosure is therefore a mean-preserving spread of the virtual type distribution, and its welfare consequences hinge entirely on the curvature of the surplus function. The central theorem characterizes this curvature precisely: total welfare improves if and only if the signal induces some consumers with negative prior virtual values to receive positive posterior virtual values—that is, if and only if some previously excluded consumers are brought into the market. When no type is excluded under the prior, Jensen’s inequality applied to the strictly concave surplus function guarantees a welfare loss. This yields a clean market-structure condition: welfare gains from data tracking require significant baseline exclusion.

The distributional consequences are equally sharp. Decomposing the welfare change into an expansion dividend and a variance term reveals a striking asymmetry: the agent bears *twice* the variance penalty relative to society as a whole, while the firm’s profit function—convex in the allocation due to the shutdown option—always generates a positive variance premium. The firm therefore gains unambiguously from signal dispersion, while the agent may lose even in markets where total welfare rises.

The remainder of the paper is organized as follows. Section 2 presents the model. Section 3 derives the martingale property of virtual valuations. Section 4 establishes the market-shutdown condition and the welfare decomposition. Section 4.3 characterizes the distributional asymmetry between agent surplus and firm profit.

2 The Model

2.1 Primitives

We study the canonical nonlinear pricing model of Mussa and Rosen (1978) and Maskin and Riley (1984), enriched with an imperfect external signal. A risk-neutral principal sells a good to a risk-neutral agent whose type $\theta \in \Theta \equiv [\underline{\theta}, \bar{\theta}] \subset \mathbb{R}_+$ is private information. The prior distribution of θ has density f and cumulative distribution F , both continuously differentiable and strictly positive on Θ .

The agent's utility from consuming quantity $q \geq 0$ and paying transfer $T \in \mathbb{R}$ is

$$U(\theta, q, T) = \theta q - T,$$

so types are indexed by marginal willingness-to-pay and utility is linear in both quantity and the transfer, ruling out income effects. The principal's cost of producing quantity q is

$$c(q) = \frac{1}{2}kq^2, \quad k > 0.$$

The quadratic specification delivers a linear first-best rule $q^*(\theta) = \theta/k$ and ensures that total surplus $S(\theta, q) \equiv \theta q - \frac{1}{2}kq^2$ is strictly concave in q for every θ — the curvature property on which the welfare analysis turns.

2.2 Information Structure

Before contracting, an exogenous signal $\sigma \in \Sigma$ about the agent's type is publicly observed by the principal. In this paper, we assume that the signal is generated by environment — telematics data, credit records, or similar external sources. Formally, it is described by a conditional density $\pi(\sigma \mid \theta)$, which together with the prior $f(\theta)$ induces a joint density

$$h(\theta, \sigma) = \pi(\sigma \mid \theta) f(\theta)$$

on $\Theta \times \Sigma$. The marginal density of σ is $g(\sigma) = \int_{\Theta} h(\theta, \sigma) d\theta$, and Bayes' rule delivers the posterior

$$f(\theta \mid \sigma) = \frac{h(\theta, \sigma)}{g(\sigma)}, \quad F(\theta \mid \sigma) = \int_{\underline{\theta}}^{\theta} f(t \mid \sigma) dt.$$

Two conditions are imposed on the information structure.

Assumption 2.1 (MLRP). The joint density $h(\theta, \sigma)$ satisfies the strict monotone likelihood ratio property: for all $\theta_1 > \theta_2$ and $\sigma_1 > \sigma_2$,

$$\frac{h(\theta_1, \sigma_1)}{h(\theta_1, \sigma_2)} > \frac{h(\theta_2, \sigma_1)}{h(\theta_2, \sigma_2)}.$$

MLRP implies that a higher signal is good news about the agent's type in the likelihood ratio order. It preserves the monotonicity of the screening problem across signal realizations: conditional on any σ , higher types remain more profitable to serve.

Assumption 2.2 (Regularity). For each $\sigma \in \Sigma$, the posterior $f(\theta \mid \sigma)$ is strictly log-concave in θ . Equivalently, the conditional inverse hazard rate

$$H(\theta \mid \sigma) \equiv \frac{1 - F(\theta \mid \sigma)}{f(\theta \mid \sigma)} \tag{1}$$

is strictly decreasing in θ .

Regularity is the standard single-crossing condition on the information rent. It ensures that the posterior virtual valuation $v^\sigma(\theta) \equiv \theta - H(\theta \mid \sigma)$ is strictly increasing in θ for every σ , so the principal's problem is globally well-behaved and the optimal allocation rule is free of bunching.

2.3 Timing

1. The information structure $\pi(\sigma \mid \theta)$ is fixed and commonly known.
2. Nature draws $\theta \sim F$; the signal $\sigma \sim \pi(\cdot \mid \theta)$ is realized and observed by the principal.
3. The principal offers a direct-revelation menu $\{(q(\hat{\theta}), T(\hat{\theta}))\}_{\hat{\theta} \in \Theta}$ to maximize expected profit given the posterior $f(\theta \mid \sigma)$.
4. The agent privately observes θ , selects a contract from the menu, and payoffs are realized.

The timing makes explicit the source of the residual screening problem. After σ is realized, the agent privately observes θ while the principal knows only the posterior $f(\theta \mid \sigma)$. Because the signal is imperfect, this posterior has non-degenerate support, and the principal must still leave informational rents to high types to prevent mimicry. It is precisely this residual rent — the standard second-degree distortion, now conditioned on σ — that drives the welfare analysis of the paper.

3 Screening and Virtual Valuations

3.1 The Optimal Contract

Given a signal realization σ , the principal faces a standard screening problem against the posterior $f(\theta \mid \sigma)$ derived in Section 2. By the revelation principle (Myerson, 1981), attention restricts without loss of generality to direct-revelation mechanisms $(q(\theta), T(\theta))_{\theta \in \Theta}$ satisfying incentive compatibility (IC) and individual rationality (IR). Standard integration by parts reduces the principal's problem to maximizing expected *virtual surplus* subject to a monotonicity constraint on q and the non-negativity constraint $q \geq 0$.

Define the conditional inverse hazard rate $H(\theta \mid \sigma)$ as in (1), and the corresponding *posterior virtual valuation*

$$v^\sigma(\theta) \equiv \theta - H(\theta \mid \sigma). \quad (2)$$

This is the signal-adjusted version of Myerson (1981)'s virtual type: it discounts the agent's marginal value θ by the information rent the principal must concede to prevent mimicry by higher types. Assumption 2.2 ensures $v^\sigma(\theta)$ is strictly increasing in θ for every σ , so the monotonicity constraint is non-binding and the optimal allocation solves the pointwise problem

$$\max_{q \geq 0} [v^\sigma(\theta) - kq] q,$$

yielding the closed-form rule

$$q^\sigma(\theta) = \max \left\{ 0, \frac{v^\sigma(\theta)}{k} \right\}. \quad (3)$$

The principal serves type θ under signal σ if and only if $v^\sigma(\theta) > 0$; otherwise the market shuts down for that type–signal pair. For later reference, define the *no-signal benchmark* as the allocation the principal would choose using only the prior $f(\theta)$:

$$v^\phi(\theta) \equiv \theta - H^\phi(\theta), \quad H^\phi(\theta) \equiv \frac{1 - F(\theta)}{f(\theta)}, \quad q^\phi(\theta) = \max \left\{ 0, \frac{v^\phi(\theta)}{k} \right\}. \quad (4)$$

Throughout, the superscript ϕ denotes variables in the no-signal benchmark and σ denotes variables conditional on a realized signal.

3.2 The Martingale Property

The key structural result of this section is that the signal induces a *mean-preserving spread* of virtual valuations across signal realizations.

Lemma 3.1 (Martingale Property). *Under Assumptions 2.1 and 2.2, for every $\theta \in \Theta$,*

$$\mathbb{E}[H(\theta | \sigma) | \theta] = H^\phi(\theta), \quad (5)$$

or equivalently, $\mathbb{E}[v^\sigma(\theta) | \theta] = v^\phi(\theta)$.

Proof. Fix θ . By Bayes' rule, $f(\sigma | \theta) = h(\theta, \sigma)/f(\theta)$. Hence

$$\mathbb{E}[H(\theta | \sigma) | \theta] = \frac{1}{f(\theta)} \int_{\Sigma} H(\theta | \sigma) f(\theta | \sigma) g(\sigma) d\sigma.$$

Since $H(\theta | \sigma) f(\theta | \sigma) = 1 - F(\theta | \sigma)$ by definition of H , and the law of iterated expectations gives $\int_{\Sigma} F(\theta | \sigma) g(\sigma) d\sigma = F(\theta)$, we obtain

$$\mathbb{E}[H(\theta | \sigma) | \theta] = \frac{1 - F(\theta)}{f(\theta)} = H^\phi(\theta).$$

The statement for virtual values follows immediately from linearity: $v^\sigma(\theta) = \theta - H(\theta | \sigma)$. $\square$

Lemma 3.1 has a transparent economic interpretation. The information structure $\pi(\sigma | \theta)$ redistributes probability mass across posteriors but, by Bayes' law, cannot alter the expected posterior distribution—a property widely formalized in the information design literature as Bayesian plausibility (Kamenica and Gentzkow, 2011). Because the information rent $H(\theta | \sigma)$ is a function of the posterior alone, its cross-signal expectation must equal its prior value. This specific martingale property of virtual valuations mirrors the fundamental observation in Eső and Szentes (2007), though applied here to exogenous signals observed by the principal rather than endogenous disclosure to the agent. Equivalently, the signal is a mean-preserving spread of the virtual type distribution. It raises the dispersion of $v^\sigma(\theta)$ around its mean $v^\phi(\theta)$ without shifting that mean.

This observation is the analytical foundation for all welfare results that follow. Since the optimal allocation (3) is a convex function of the virtual value, Jensen's inequality immediately implies that signal dispersion weakly *expands* expected output for every type. Whether this output expansion improves welfare depends on where in the type distribution the expansion occurs — the question taken up in Section 4.

4 Welfare Analysis

This section characterizes the welfare consequences of the signal. We compare the signaling regime—in which the principal observes σ and screens against the posterior $f(\theta | \sigma)$ —against the no-signal benchmark defined in (4). Define the pointwise welfare change

$$\Delta W(\theta) \equiv \mathbb{E}_{\sigma|\theta}[S(\theta, q^\sigma(\theta))] - S(\theta, q^\phi(\theta)), \quad (6)$$

so that the aggregate welfare change is $\Delta TW = \int_{\Theta} \Delta W(\theta) f(\theta) d\theta$.

4.1 The Market-Shutdown Condition

The martingale property (Lemma 3.1) establishes that the signal is a mean-preserving spread of virtual valuations across signal realizations. The welfare consequences of this spread depend critically on the shape of the indirect surplus function. When every consumer is served under every signal realization, the allocation is everywhere interior and surplus is globally concave in the virtual value—a mean-preserving spread then reduces welfare by Jensen’s inequality. The non-negativity constraint on quantity is what opens the door to welfare gains: if the signal is dispersed enough to drive some posterior virtual values below zero, the principal shuts down the market for those type–signal pairs, and the truncation at zero introduces a locally convex region that can overturn the Jensen penalty.

Lemma 4.1 (No welfare gain without market shutdown). *Suppose Assumptions 2.1 and 2.2 hold and the information structure is non-degenerate. If $q^\sigma(\underline{\theta}) > 0$ for all $\sigma \in \Sigma$, then*

$$\Delta TW \leq 0,$$

with strict inequality whenever $\text{Var}[v^\sigma(\theta) \mid \theta] > 0$ on a set of positive measure.

Proof. The premise $q^\sigma(\underline{\theta}) > 0$ requires $v^\sigma(\underline{\theta}) > 0$ for all $\sigma \in \Sigma$. By Assumption 2.2, $v^\sigma(\theta)$ is strictly increasing in θ , so $v^\sigma(\theta) > 0$ for all $(\theta, \sigma) \in \Theta \times \Sigma$. Global positivity implies the non-negativity constraint is everywhere slack, and the optimal allocation is always the interior solution $q^\sigma(\theta) = v^\sigma(\theta)/k$.

Substituting into $S(\theta, q) = \theta q - \frac{1}{2}kq^2$, the indirect welfare function is

$$W^{\text{ind}}(v^\sigma; \theta) = S\left(\theta, \frac{v^\sigma}{k}\right) = \frac{1}{k}\left(\theta v^\sigma - \frac{1}{2}(v^\sigma)^2\right).$$

This is strictly concave in v^σ , since $\partial^2 W^{\text{ind}}/\partial (v^\sigma)^2 = -1/k < 0$.

The martingale property (Lemma 3.1) gives $\mathbb{E}_{\sigma|\theta}[v^\sigma(\theta)] = v^\phi(\theta)$ for every θ . Applying Jensen’s inequality to the strictly concave function $W^{\text{ind}}(\cdot; \theta)$:

$$\mathbb{E}_{\sigma|\theta}\left[W^{\text{ind}}(v^\sigma(\theta); \theta)\right] \leq W^{\text{ind}}\left(\mathbb{E}_{\sigma|\theta}[v^\sigma(\theta)]; \theta\right) = W^{\text{ind}}(v^\phi(\theta); \theta),$$

with strict inequality when $\text{Var}[v^\sigma(\theta) \mid \theta] > 0$. Since $\Delta W(\theta) = \mathbb{E}_{\sigma|\theta}[W^{\text{ind}}(v^\sigma; \theta)] - W^{\text{ind}}(v^\phi; \theta) \leq 0$ for every θ , integrating over Θ yields $\Delta TW \leq 0$. $\square$

Remark 4.2. Lemma 4.1 restricts attention to non-degenerate information structures satisfying Assumption 2.2. A perfectly revealing signal collapses the posterior to a point mass, violates the log-concavity requirement, and eliminates the screening distortion entirely ($H(\theta \mid \sigma) \rightarrow 0$), restoring the first-best allocation $q^*(\theta) = \theta/k$. The lemma therefore establishes that, short of perfect revelation, injecting partial information strictly harms welfare unless it triggers market shutdown for some type–signal pairs.

The converse direction is equally important. The condition $q^\phi(\underline{\theta}) > 0$ —the market being active under the prior—does *not* suffice to guarantee $\Delta TW \leq 0$. Even if no type is excluded in the benchmark, a sufficiently dispersed signal can induce negative posterior virtual values for some realizations, activating the shutdown region and potentially generating a net welfare gain. The relevant condition is whether shutdown occurs *under the signal*, not under the prior.

4.2 The Welfare Decomposition

To characterize the sign and magnitude of ΔTW in the general case, we first establish that the convexity of the optimal allocation rule always produces a non-negative quantity expansion in expectation.

Lemma 4.3 (Quantity expansion). *For any $\theta \in \Theta$, define the expected quantity expansion*

$$\Delta \bar{q}(\theta) \equiv \mathbb{E}_{\sigma|\theta}[q^\sigma(\theta)] - q^\phi(\theta).$$

Then $\Delta \bar{q}(\theta) \geq 0$ for all $\theta \in \Theta$, with strict inequality if and only if the support of $v^\sigma(\theta)$ straddles zero.

Proof. The optimal allocation rule (3) can be written as $q(v) = \max\{0, v/k\}$, which is a convex function of v . The martingale property gives $\mathbb{E}_{\sigma|\theta}[v^\sigma(\theta)] = v^\phi(\theta)$. Applying Jensen's inequality to the convex function $q(\cdot)$:

$$\mathbb{E}_{\sigma|\theta}[q^\sigma(\theta)] = \mathbb{E}_{\sigma|\theta}[q(v^\sigma(\theta))] \geq q(\mathbb{E}_{\sigma|\theta}[v^\sigma(\theta)]) = q(v^\phi(\theta)) = q^\phi(\theta).$$

The inequality is strict if and only if $q(\cdot)$ is not affine on the support of $v^\sigma(\theta)$, which occurs precisely when the support straddles the kink at $v = 0$. $\square$

Theorem 4.4 (Welfare decomposition). *Under Assumptions 2.1 and 2.2, for every $\theta \in \Theta$,*

$$\Delta W(\theta) = \underbrace{\left[\theta - \frac{k}{2} (\mathbb{E}_{\sigma|\theta}[q^\sigma(\theta)] + q^\phi(\theta)) \right]}_{\text{average social profit margin}} \cdot \underbrace{\Delta \bar{q}(\theta)}_{\text{expansion dividend}} - \underbrace{\frac{k}{2} \text{Var}[q^\sigma(\theta) | \theta]}_{\text{variance penalty}}. \quad (7)$$

Consequently, $\Delta TW > 0$ if and only if

$$\int_{\Theta} \left[\theta - \frac{k}{2} (\mathbb{E}_{\sigma|\theta}[q^\sigma] + q^\phi) \right] \Delta \bar{q}(\theta) f(\theta) d\theta > \frac{k}{2} \int_{\Theta} \text{Var}[q^\sigma | \theta] f(\theta) d\theta. \quad (8)$$

Proof. By definition (6) and the surplus function $S(\theta, q) = \theta q - \frac{k}{2} q^2$,

$$\Delta W(\theta) = \theta (\mathbb{E}_{\sigma|\theta}[q^\sigma] - q^\phi) - \frac{k}{2} (\mathbb{E}_{\sigma|\theta}[(q^\sigma)^2] - (q^\phi)^2).$$

We decompose the quadratic term using the variance identity $\mathbb{E}[(q^\sigma)^2] = \text{Var}[q^\sigma | \theta] + (\mathbb{E}[q^\sigma])^2$:

$$\begin{aligned} \mathbb{E}_{\sigma|\theta}[(q^\sigma)^2] - (q^\phi)^2 &= \text{Var}[q^\sigma | \theta] + (\mathbb{E}_{\sigma|\theta}[q^\sigma])^2 - (q^\phi)^2 \\ &= \text{Var}[q^\sigma | \theta] + \underbrace{(\mathbb{E}_{\sigma|\theta}[q^\sigma] - q^\phi)}_{\Delta \bar{q}(\theta)} (\mathbb{E}_{\sigma|\theta}[q^\sigma] + q^\phi). \end{aligned}$$

Substituting back and factoring $\Delta \bar{q}(\theta)$:

$$\begin{aligned} \Delta W(\theta) &= \theta \Delta \bar{q}(\theta) - \frac{k}{2} [\text{Var}[q^\sigma | \theta] + \Delta \bar{q}(\theta) (\mathbb{E}_{\sigma|\theta}[q^\sigma] + q^\phi)] \\ &= \left[\theta - \frac{k}{2} (\mathbb{E}_{\sigma|\theta}[q^\sigma] + q^\phi) \right] \Delta \bar{q}(\theta) - \frac{k}{2} \text{Var}[q^\sigma | \theta]. \end{aligned}$$

Integrating over Θ with weight $f(\theta)$ yields (8). $\square$

The decomposition (7) has a transparent geometric interpretation, illustrated in Figure 1 for a fixed type θ . Information disclosure raises the *expected* allocation from q^ϕ to $\mathbb{E}_{\sigma|\theta}[q^\sigma]$; the social gain from this expansion is the area of the green trapezoid bounded above by the marginal benefit θ and below by the average marginal cost $\frac{k}{2}(\mathbb{E}[q^\sigma] + q^\phi)$ —the red arrow in the figure. The variance penalty $\frac{k}{2}\text{Var}[q^\sigma | \theta]$ has no counterpart in this diagram: it reflects allocation volatility *within* the active region across signal realizations, not a shift of the boundary. The connection to Lemma 4.1 is then immediate: when no type is ever excluded, $\Delta\bar{q}(\theta) = 0$ for every θ , the trapezoid collapses to zero area, and only the variance penalty survives.

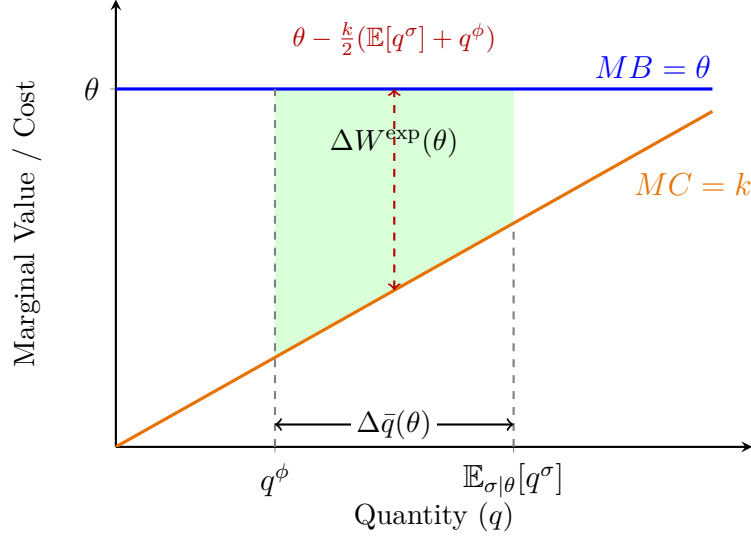


Figure 1: Geometric representation of the expansion dividend in (7) for a fixed type θ . The signal raises expected output from q^ϕ to $\mathbb{E}_{\sigma|\theta}[q^\sigma]$ by drawing marginally excluded consumers into the market. The social value of this expansion equals the area of the green trapezoid: the quantity expansion $\Delta\bar{q}(\theta)$ multiplied by the average social profit margin $\theta - \frac{k}{2}(\mathbb{E}[q^\sigma] + q^\phi)$ (red arrow). The variance penalty $\frac{k}{2}\text{Var}[q^\sigma | \theta]$ is not depicted, as it operates within the active region and carries no geometric footprint in this diagram.

Figure 2 provides numerical illustration of the welfare decomposition (7) across four specifications that vary the prior distribution and the signal structure. The green area at each type θ represents the expansion dividend; the red area represents the variance penalty. The welfare improvement condition (8) holds if and only if the aggregate green area exceeds the aggregate red area over the full type space.

4.3 Distributional Consequences

The aggregate welfare change masks a sharp asymmetry in how the signal's effects are distributed between the two parties. By standard integration by parts, the agent's expected surplus equals the expected information rent, which under the optimal contract takes the form of the *virtual agent surplus*

$$\widetilde{AS}(\theta | \sigma) \equiv H(\theta | \sigma) q^\sigma(\theta).$$

Under the optimal interior allocation, $H(\theta | \sigma) = \theta - kq^\sigma(\theta)$, so

$$\widetilde{AS}(\theta | \sigma) = \theta q^\sigma(\theta) - k(q^\sigma(\theta))^2.$$

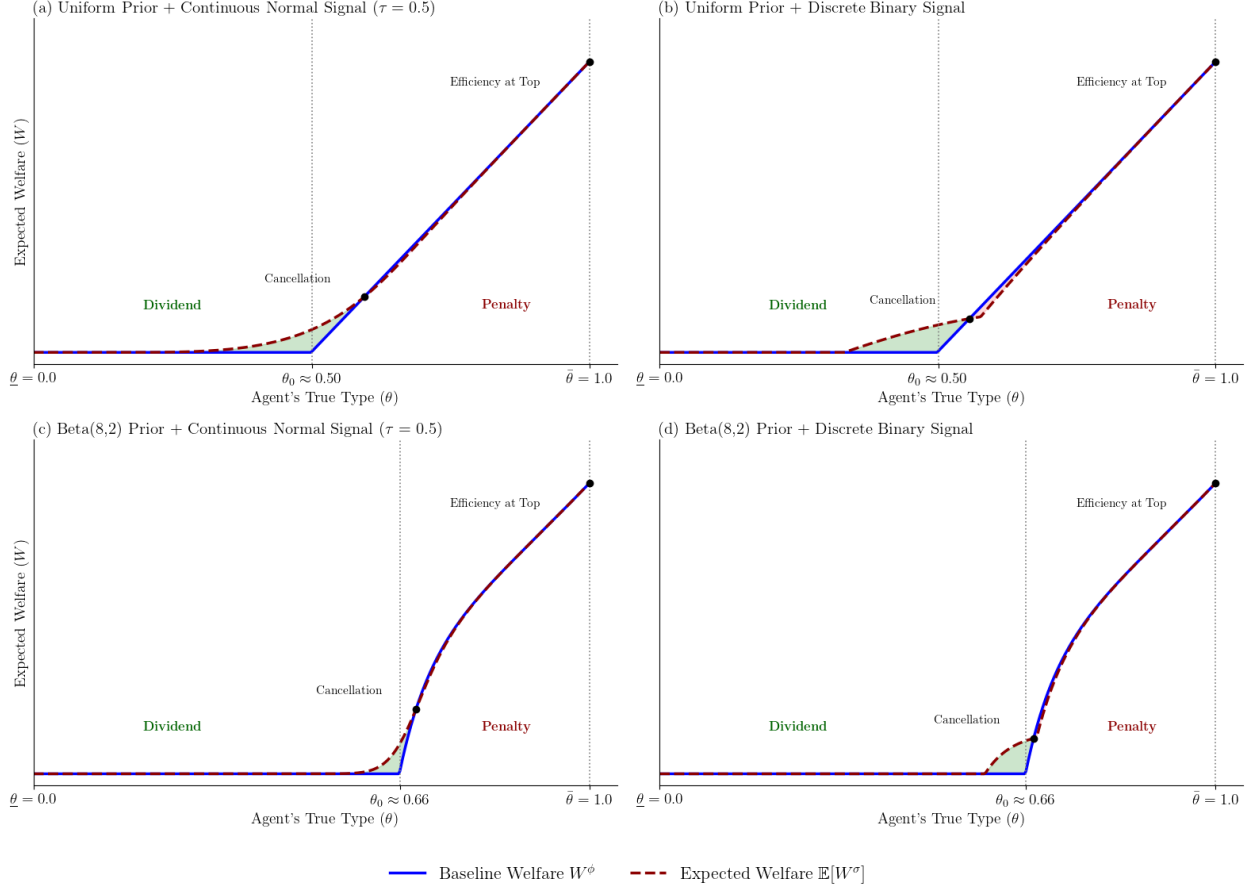


Figure 2: Numerical illustration of the welfare decomposition (7) across four specifications. Panels (a)–(b) use a uniform prior; panels (c)–(d) use a right-skewed Beta(8,2) prior, which reveals the strict concavity of the welfare function suppressed by the algebraic cancellation under uniformity. The left column uses continuous Gaussian noise ($\tau = 0.5$); the right column uses a discrete binary signal. In all panels, the expansion dividend (green) dominates near $\underline{\theta}$ where market shutdown is most likely to be triggered by the signal, while the variance penalty (red) dominates near $\bar{\theta}$ where all types are always served. The four panels together confirm that the market-shutdown condition of Lemma 4.1 is the binding constraint on welfare improvement: welfare gains are confined to the lower region of the type space where baseline exclusion is present.

The principal's virtual profit is the residual

$$\tilde{\Pi}(\theta \mid \sigma) \equiv S(\theta, q^\sigma) - \widetilde{AS}(\theta \mid \sigma) = \frac{k}{2}(q^\sigma(\theta))^2,$$

which is strictly convex in the allocation.

Corollary 4.5 (Distributional decomposition). *Under Assumptions 2.1 and 2.2, the pointwise changes in agent surplus and firm profit satisfy*

$$\Delta \widetilde{AS}(\theta) = \left[\theta - k(\mathbb{E}_{\sigma|\theta}[q^\sigma] + q^\phi) \right] \Delta \bar{q}(\theta) - k \text{Var}[q^\sigma(\theta) \mid \theta], \quad (9)$$

$$\Delta \tilde{\Pi}(\theta) = \frac{k}{2}(\mathbb{E}_{\sigma|\theta}[q^\sigma] + q^\phi) \Delta \bar{q}(\theta) + \frac{k}{2} \text{Var}[q^\sigma(\theta) \mid \theta]. \quad (10)$$

Proof. For (9), compute

$$\Delta \widetilde{AS}(\theta) = \mathbb{E}_{\sigma|\theta}[\theta q^\sigma - k(q^\sigma)^2] - (\theta q^\phi - k(q^\phi)^2).$$

Separating the linear and quadratic terms:

$$\Delta \widetilde{AS}(\theta) = \theta \Delta \bar{q}(\theta) - k(\mathbb{E}_{\sigma|\theta}[(q^\sigma)^2] - (q^\phi)^2).$$

Applying the same variance decomposition as in the proof of Theorem 4.4,

$$\mathbb{E}_{\sigma|\theta}[(q^\sigma)^2] - (q^\phi)^2 = \text{Var}[q^\sigma \mid \theta] + \Delta \bar{q}(\theta)(\mathbb{E}_{\sigma|\theta}[q^\sigma] + q^\phi),$$

and factoring $\Delta \bar{q}(\theta)$ yields (9).

For (10), compute

$$\Delta \tilde{\Pi}(\theta) = \frac{k}{2}(\mathbb{E}_{\sigma|\theta}[(q^\sigma)^2] - (q^\phi)^2) = \frac{k}{2}[\text{Var}[q^\sigma \mid \theta] + \Delta \bar{q}(\theta)(\mathbb{E}_{\sigma|\theta}[q^\sigma] + q^\phi)],$$

which rearranges to (10). Internal consistency is verified by noting that (9) + (10) = (7). $\square$

Comparing (9) with (7) reveals the asymmetry precisely. The variance coefficient in the agent's decomposition is k —exactly *twice* the social variance penalty $\frac{k}{2}$. The agent therefore absorbs twice the second-order cost of allocation volatility relative to the social optimum. The firm, by contrast, earns a strict *variance premium* $+\frac{k}{2}\text{Var}[q^\sigma \mid \theta] > 0$ in (10): because the principal's downside is truncated at zero by the shutdown option, the profit function $\tilde{\Pi} \propto q^2$ is globally convex in the allocation, and Jensen's inequality works in the firm's favor. The firm therefore gains unambiguously from signal dispersion, while the agent may lose even when total welfare rises. Data tracking can be simultaneously welfare-improving for the social planner and welfare-reducing for the average consumer.

5 Conclusion

This paper has studied the welfare consequences of imperfect external signals in a nonlinear pricing environment—a regime we term 2.5-degree price discrimination, lying between the classical second- and third-degree benchmarks. The central analytical object is the posterior virtual valuation, which inherits a martingale property from Bayes' law: for every type θ , the prior virtual value equals the expectation of its posterior counterpart across all signal realizations. This establishes that

information disclosure is a mean-preserving spread of the virtual type distribution, and all welfare results follow from the interplay between this spread and the curvature of the surplus function.

Three findings emerge from the analysis. First, the market-shutdown condition (Lemma 4.1) identifies the precise structural prerequisite for welfare gains: total welfare can improve if and only if the signal induces market shutdown for some type–signal pairs, drawing previously excluded consumers into the market. When every type is served under every signal realization, the surplus function is globally concave and Jensen’s inequality guarantees a welfare loss. Second, the welfare decomposition (Theorem 4.4) separates the aggregate welfare change into an expansion dividend—the trapezoidal surplus generated by newly-served consumers—and a variance penalty from allocation volatility. Whether the signal improves or reduces welfare hinges entirely on which term dominates. Third, the distributional decomposition (Corollary 4.5) reveals a structural asymmetry: the agent bears twice the variance penalty relative to the social optimum, while the firm earns an unambiguous variance premium because its profit function is globally convex in the allocation. Data tracking therefore reallocates surplus from consumers to the firm even in markets where it raises total welfare.

Directions for Future Research

The model is deliberately parsimonious, and several extensions are both natural and economically important.

Optimal information design. The most immediate extension is to endogenize the signal. Throughout this paper, the information structure $\pi(\sigma \mid \theta)$ is treated as exogenous—generated by the environment rather than by either party. A natural next step is to ask which party would design the signal if given the choice, and what information structure each would choose.

From the firm’s perspective, the distributional decomposition shows that the principal gains unambiguously from any mean-preserving spread of virtual values. This suggests the firm would prefer the most dispersed signal consistent with the regularity assumption—one that maximizes the variance premium while preserving the monotonicity of the optimal allocation. Characterizing this optimal signal for the principal is a well-posed mechanism design problem, closely related to the persuasion framework of Kamenica and Gentzkow (2011): the principal chooses a signal to maximize its own payoff, taking the agent’s optimal response as given.

From the agent’s perspective, the picture is more subtle. The agent bears twice the variance penalty but also benefits from the expansion dividend when it is positive. Whether the agent would prefer a more or less informative signal depends on the market structure—specifically, on whether baseline exclusion is present and how large the expansion dividend is relative to the variance cost. A social planner maximizing total welfare faces a similar trade-off, with the additional consideration that the expansion dividend and variance penalty enter with different weights than they do for either individual party.

Formalizing these problems requires specifying a feasible set of information structures—likely those satisfying MLRP and regularity—and solving a functional optimization problem over conditional densities $\pi(\sigma \mid \theta)$. The martingale property established in Lemma 3.1 provides a useful constraint: any feasible signal must preserve the mean of the virtual type distribution, so the design problem reduces to choosing the *dispersion* of the spread subject to regularity. This is a tractable direction that the current analysis brings within reach.

Bilateral private information. The present model assumes the signal is publicly observed by the principal alone after the agent’s type is realized. A richer setting would allow the agent to receive

a private signal about their own type before contracting—as in the framework of Eső and Szentes (2007)—generating bilateral asymmetric information. In such a setting, the agent’s information rent depends on both the principal’s posterior and the agent’s private belief, and the interaction between the two information structures can generate qualitatively new effects on screening and welfare.

Competition and market structure. The welfare results derived here apply to a monopolist. Extending the analysis to competing firms would introduce a second margin: imperfect signals not only affect the screening distortion within each firm, but also shape how consumers sort across firms. Whether competition amplifies or dampens the welfare effects of data tracking is an open question with direct policy relevance, particularly in the context of platform markets where firms share access to consumer data through common intermediaries such as data brokers.

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